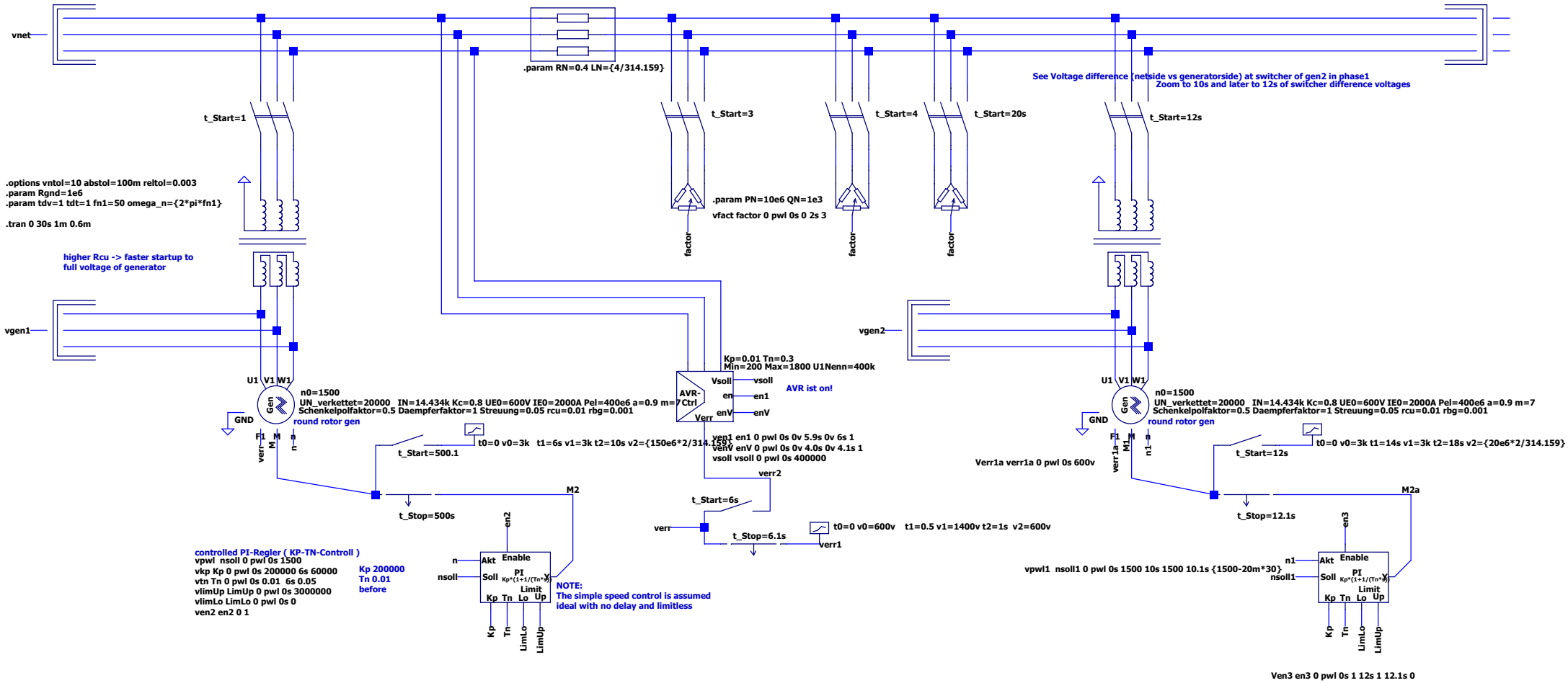
vnet
.param RN=0.4 LN={4/314.159}
See Voltage difference (netside vs generatorside) at switcher of gen2 in phase1
Zoom to 10s and later to 12s of switcher difference voltages
t_Start=1
t_Start=3
t_Start=4
t_Start=20s
t_Start=12s
.options vntol=10 abstol=100m reltol=0.003
.param Rgnd=1e6
.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1}
.tran 0 30s 1m 0.6m
.param PN=10e6 QN=1e3
vfact factor 0 pwl 0s 0 2s 3
factor
factor
factor
higher Rcu -> faster startup to
full voltage of generator
vgen1
vgen2
Kp=0.01 Tn=0.3
Min=200 Max=1800 U1Nenn=400k
AVR ist on!
U1 V1 W1
Gen
n0=1500
UN_verkettet=20000 IN=14.434k Kc=0.8 UE0=600V IE0=2000A Pel=400e6 a=0.9 m=7
Schenkelpolfaktor=0.5 Daempferfaktor=1 Streuung=0.05 rcu=0.01 rbg=0.001
round rotor gen
GND
AVR-Ctrl
Vsoll
en
enV
Verr
vsoll
en1
enV
F1 M n
verr M n
t_Start=500.1
t0=0 v0=3k t1=6s v1=3k t2=10s v2={150e6*2/314.159}
ven1 en1 0 pwl 0s 0v 5.9s 0v 6s 1
venV enV 0 pwl 0s 0v 4.0s 0v 4.1s 1
vsoll vsoll 0 pwl 0s 400000
verr2
Verr1a verr1a 0 pwl 0s 600v
t_Start=12s
t0=0 v0=3k t1=14s v1=3k t2=18s v2={20e6*2/314.159}
verr1a M1 n1
M2
M2a
t_Stop=500s
t_Stop=12.1s
t_Start=6s
verr
t_Stop=6.1s
t0=0 v0=600v t1=0.5 v1=1400v t2=1s v2=600v
verr1
controlled PI-Regler (KP-TN-Controll)
vpwl nsoll 0 pwl 0s 1500
vkp Kp 0 pwl 0s 200000 6s 60000
vtn Tn 0 pwl 0s 0.01 6s 0.05
vlimUp LimUp 0 pwl 0s 3000000
vlimLo LimLo 0 pwl 0s 0
ven2 en2 0 1
Kp 200000
Tn 0.01
before
en2
n
nsoll
Akt
Soll
Enable
PI
Kp*(1+1/(Tn*s))
Limit
Kp Tn Lo Up
NOTE:
The simple speed control is assumed
ideal with no delay and limitless
vpwl1 nsoll1 0 pwl 0s 1500 10s 1500 10.1s {1500-20m*30}
en3
n1
nsoll1
Ven3 en3 0 pwl 0s 1 12s 1 12.1s 0